INTERACTIVE PLAYER EVALUATION DASHBOARD FOR NCAA DIVISION I HOCKEY

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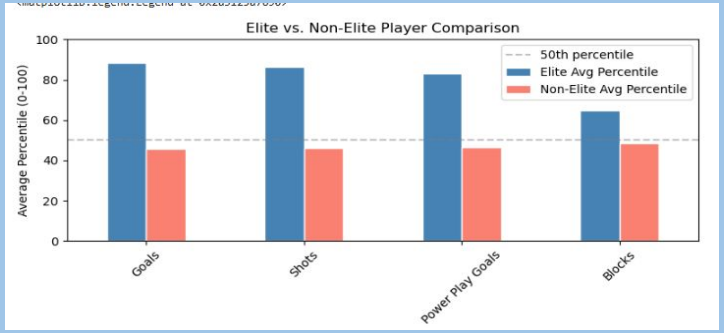
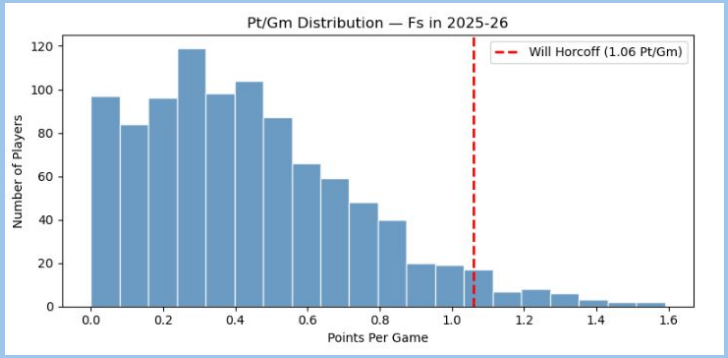
SAL 603 Final Project

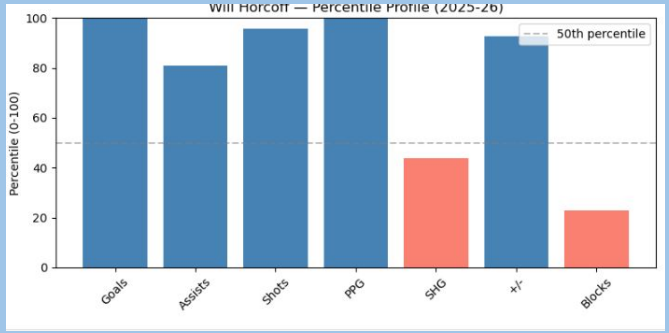
Project Objective

- ➔ Use Python to analyze Division I hockey player statistics
- ➔ Compare player performance relative to peers
- ➔ Identify statistical patterns among high-performing players

📊 DATA
5 Seasons of NCAA Player Stats
↓
🐍 PYTHON
Data Cleaning + Transformation
↓
📈 ANALYSIS
Percentiles + Visualizations
↓
💡 INSIGHTS
Patterns in Elite Player Performance







Research Question

How does a Division I hockey player's statistical profile compare to same-position peers within each season, and how does that profile evolve across a five-year eligibility window?

Dataset Overview

Dataset includes five NCAA Division I seasons

- 2021–22
- 2022–23
- 2023–24
- 2024–25
- 2025–26

Player statistics collected

- Games Played
- Goals
- Assists
- Points
- Shots
- Power Play Goals
- Blocks
- Faceoff Percentage

Data Sources

- Player statistics collected from CollegeHockeyInc.com
- Data was transformed into a CSV in Excel
- Data then imported into Python using pandas

```
[9]: # Load each season's data
     # Data from CollegeHockeyInc.com
     stats_2022 = pd.read_csv("college_hockey_stats_2021_22.csv")
     stats_2023 = pd.read_csv("college_hockey_stats_2022_23.csv")
     stats_2024 = pd.read_csv("college_hockey_stats_2023_24.csv")
     stats_2025 = pd.read_csv("college_hockey_stats_2024_25.csv")
     stats_2026 = pd.read_csv("college_hockey_stats_2025_26.csv")

     # Tag each dataframe with its season before combining
     stats_2022["Season"] = "2021-22"
     stats_2023["Season"] = "2022-23"
     stats_2024["Season"] = "2023-24"
     stats_2025["Season"] = "2024-25"
     stats_2026["Season"] = "2025-26"

     # Combine all five seasons into one dataframe
     five_year_stats = pd.concat([stats_2022, stats_2023, stats_2024, stats_2025, stats_2026], ignore_index=True)
```

Data Cleaning Process

1. Combined five seasons into one dataset
2. Removed unused columns
3. Standardized position labels
4. Created forward and defense position groups

```python
# Combine all five seasons into one dataframe
five_year_stats = pd.concat([stats_2022, stats_2023, stats_2024, stats_2025, stats_2026], ignore_index=True)

# Drop the rank column and any unnamed columns that come from the CSV index
five_year_stats = five_year_stats.drop(columns=["Rk"], errors="ignore")
five_year_stats = five_year_stats.loc[:, ~five_year_stats.columns.str.contains("^Unnamed")]

five_year_stats
```

[9]:

	Name	Team	Pos.	Yr	GP	G	A	PTS	Pt/Gm	PIM	Shots	GWG	PPG	SHG	+/-	Blk	FW	FL	FO%	Season
0	Bobby Brink	Denver	F	Jr	41.0	14.0	43.0	57.0	1.3902	44.0	119.0	2.0	5.0	1.0	27.0	17.0	2.0	4.0	33.3	2021-22
1	Nathan Smith	Minnesota State	F	Jr	38.0	19.0	31.0	50.0	1.3158	43.0	101.0	4.0	3.0	0.0	22.0	11.0	329.0	243.0	57.5	2021-22
2	Julian Napravnik	Minnesota State	F	Sr	40.0	18.0	31.0	49.0	1.2250	10.0	129.0	3.0	6.0	0.0	32.0	10.0	0.0	0.0	NaN	2021-22
3	Bobby Trivigno	Massachusetts	F	Sr	37.0	20.0	29.0	49.0	1.3243	28.0	126.0	4.0	4.0	1.0	22.0	14.0	2.0	4.0	33.3	2021-22
4	Owen Sillinger	Bemidji State	F	Sr	39.0	17.0	30.0	47.0	1.2051	41.0	137.0	3.0	2.0	0.0	10.0	15.0	515.0	425.0	54.8	2021-22
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

Data Transformation

**To compare players fairly, percentile rankings were calculated.*

Percentiles were computed by:

- Season
- Position group (Forwards vs Defensemen)
- (A forward in 2023–24 is compared only to other forwards in 2023–24)

Metrics converted to percentiles:

- Points per Game
- Goals
- Assists
- Shots
- Power Play Goals
- Blocks
- Faceoff Percentage

[48]:
```python
# Function to convert a stat into a 0-100 percentile rank
def percentile_rank(series):
    return series.rank(pct=True) * 100

# Calculate Pt/Gm percentile within each season and position group
five_year_stats["PtGm_pct"] = five_year_stats.groupby(["Season", "Pos_Group"])["Pt/Gm"].transform(percentile_rank)

# List of stats to percentile rank
percentile_metrics = ["G", "A", "Shots", "PPG", "SHG", "+/-", "Blk", "FO%"]

# Loop through each stat and create a percentile column
# FO% becomes FO_pct, everything else just gets _pct added
for metric in percentile_metrics:
    pct_col = metric.replace("%", "") + "_pct"
    five_year_stats[pct_col] = five_year_stats.groupby(["Season", "Pos_Group"])[metric].transform(percentile_rank)

# Quick check of the results
five_year_stats[["Name", "Season", "Pos_Group", "G", "G_pct", "Shots", "Shots_pct"]].head(10)
```

[48]:

	Name	Season	Pos_Group	G	G_pct	Shots	Shots_pct
0	Bobby Brink	2021-22	F	14.0	93.756558	119.0	97.691501
1	Nathan Smith	2021-22	F	19.0	98.216159	101.0	93.966422
2	Julian Napravnik	2021-22	F	18.0	97.743966	129.0	98.530955
3	Bobby Trivigno	2021-22	F	20.0	98.740818	126.0	98.216159

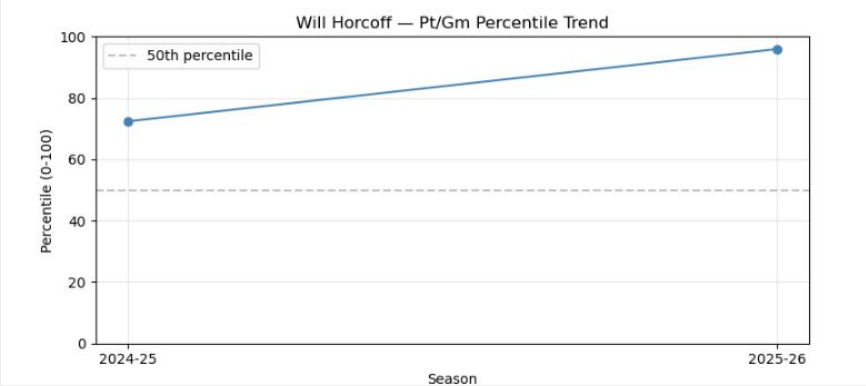
Player Performance Trends Across Seasons

Line chart showing

- player performance relative to peers
- development across seasons

This visualization helps identify

- improvement
- regression
- consistent performance

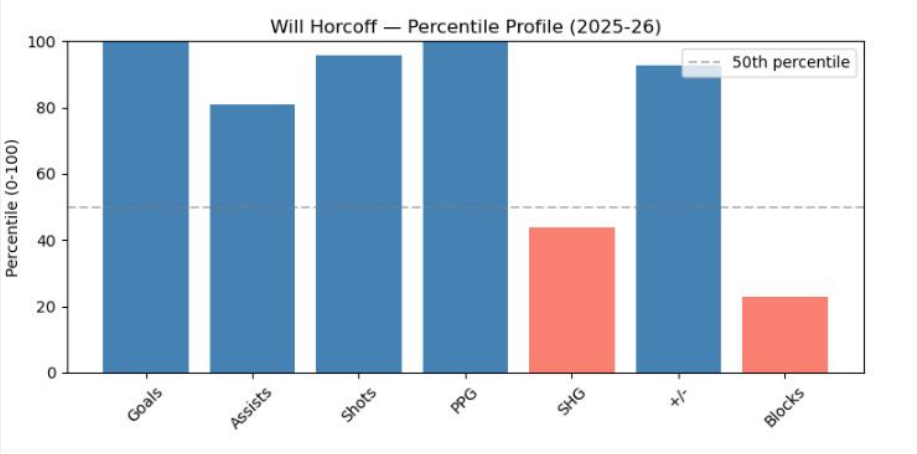


Player Statistical Profile

**This chart shows how a player ranks across multiple metrics

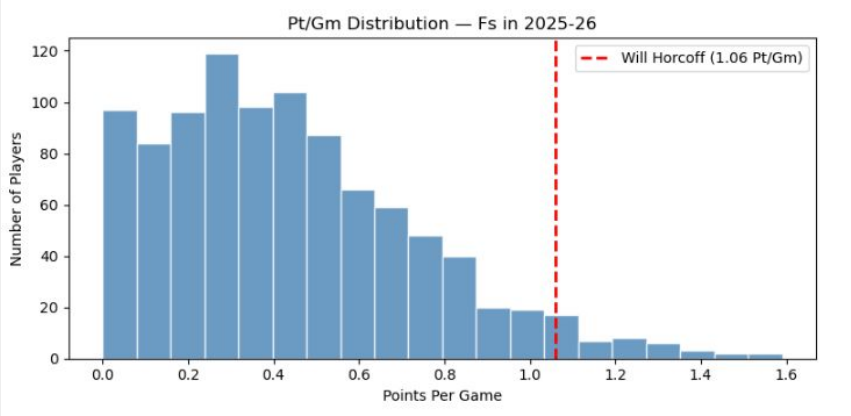
Example metrics

- ★ Goals percentile
- ★ Assists percentile
- ★ Shot volume percentile
- ★ Power play production
- ★ Defensive blocks



Player Distribution Within Position Group

- Histogram showing distribution of Points per Game for all players
- Player value highlighted in the distribution
- This confirms percentile rankings visually



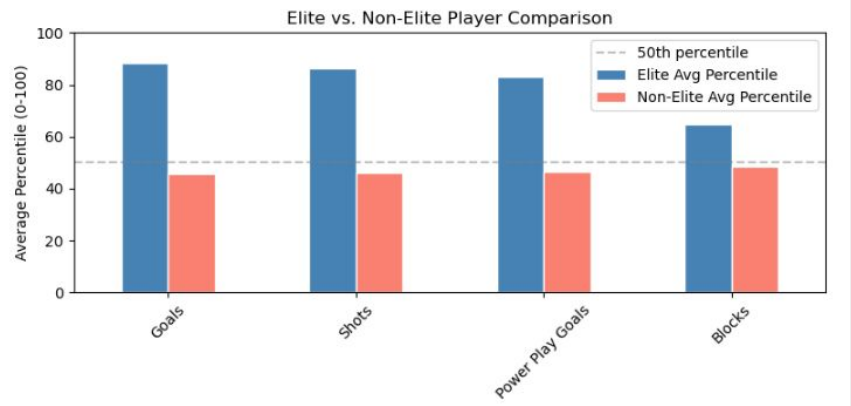
Statistical Comparison

Players were grouped into

- **Elite players**
  - Top 10% in Points per Game percentile
- **Non-elite players**
  - Remaining players

Average percentile comparisons show

- Elite players shoot significantly more
- Elite players score more goals
- Power play scoring is also higher among elite players



Key Findings

🎯 Shot volume strongly differentiates elite players

🥅 Elite players consistently rank high in multiple offensive metrics

🛑 Defensive metrics show smaller differences between groups

📈 Percentile rankings provide meaningful context for evaluating players

Project Summary

This project demonstrated how Python can be used to :

- ➔ combine multiple datasets
- ➔ clean and transform sports statistics
- ➔ analyze player performance
- ➔ create meaningful visualizations

The analysis provides insight into how NCAA hockey players compare to their peers.

Thank you